

Reconstructing a Number from its Two Largest Proper Divisors

Demonstrating Algorithmic Intuition and Structured Communication in Engineering Interviews

In a technical interview, solving the problem is only half the battle. Interviewers are rarely tracking just the final formula; instead, they value your **thought process**—how you navigate constraints, deduce mathematical principles from basic patterns, and cleanly articulate an optimized solution. Below is a structured, step-by-step framework designed to communicate the optimal solution for reconstructing an unknown number from its two largest proper divisors.

The 6-Step Derivation Framework

Step 1: Restate the Problem Clear and Intentionally

Begin by explicitly clarifying the input parameters and the ultimate objective. This establishes baseline alignment with your interviewer and confirms you understand the definitions at play.

"We are given the two largest proper divisors, a and b , of an unknown number x . Our goal is to reconstruct the smallest possible valid integer value for x ."

Step 2: State the Foundational Mathematical Observation

Build the bridge from the definition of a divisor to the concept of multiples. By definition, if a and b are divisors of x , then x must simultaneously be an exact multiple of both numbers.

"Since both a and b are proper divisors of x , it follows that x must be a common multiple of both a and b . Because we want to minimize x , the most logical initial candidate to investigate is their Least Common Multiple (LCM)."

Step 3: Establish the Base Case (Why LCM is the Primary Candidate)

Walk through a concrete example where the LCM maps perfectly to the correct answer. This shows that your intuition holds true under standard conditions.

SCENARIO A: COPRIME OR MULTI-FACTOR COMPOSITIONS

Inputs:	$a = 4, b = 6$
Computed LCM:	12
Divisors of 12:	1, 2, 3, 4, 6, 12 (Proper divisors: 1, 2, 3, 4, 6)
Verification:	The two largest proper divisors are exactly 6 and 4.

"As we can see, the LCM provides the smallest possible value that naturally satisfies the divisibility rules for both inputs simultaneously."

Step 4: Analyze Edge Cases and Identify Failure Modes

Demonstrate critical thinking by proactively exploring where the initial LCM intuition fails. Specifically, highlight what happens when one input cleanly divides the other.

SCENARIO B: DIRECT DIVISIBILITY CONSTRAINT VIOLATION

Inputs:	$a = 3, b = 9$
Computed LCM:	9
The Issue:	x cannot equal 9 because a number cannot be a *proper* divisor of itself.

"If the LCM equals b (which occurs when a divides b), b ceases to be a proper divisor; it becomes the number itself. This directly violates the problem constraints, requiring us to scale the candidate upward."

Step 5: Formulate the Scaling Fix

Explain how to mathematically scale the number minimalistically without adding intermediate divisors that shift the positions of a and b .

If we try multiplying b by 2 when $b = 9$ and $a = 3$, we get 18. The divisors of 18 are 1, 2, 3, 6, 9, 18—meaning the second largest proper divisor shifts to 6 instead of 3. To maintain the structural ratio guaranteed by a valid input, we must multiply by the ratio of the two factors:

$$\text{Factor} = b / a \rightarrow x = b \times (b / a)$$

For our failure case: $9 \times (9 / 3) = 27$. The proper divisors of 27 are 1, 3, and 9. This perfectly restores 9 and 3 as the top two proper divisors.

Step 6: Synthesize the Final Generalized Algorithm

State the absolute, clean programmatic logical flow based on your deductions:

1. Compute the Least Common Multiple (LCM) of a and b .
2. If $LCM > b$, return LCM .
3. Otherwise (when $LCM == b$), return $b \times (b / a)$.

Handling Direct Follow-Up Questions

Q: Why start with the LCM?

Answer: Because x must be divisible by both parameters, it is geometrically bounded to be a common multiple. Since the problem demands the smallest valid version of x , checking the baseline least common multiple is the mathematically optimal starting candidate.

Q: Why exactly does the formula $b \times (b / a)$ work when a divides b ?

Answer: When a divides b , the LCM matches b , which is invalid. To preserve the relative scale and ensure no other arbitrary divisors slip between a and b , we must expand b by the relative factor difference between the two known values, which is b / a . Given the problem guarantees a valid configuration exists, this yields the exact minimal bounding target.

The "Elevator Pitch" Interview Script (2-Minute Summary)

If you need to deliver a fast, high-impact version of this entire breakdown during a live coding session, use this phrasing:

"Since both numbers are proper divisors of the required target, the result must inherently be a common multiple. To secure the smallest possible configuration, I first evaluate the LCM. If the LCM is strictly greater than the larger divisor b , then b remains intact as the largest proper divisor, rendering the LCM our final answer.

The unique edge case happens when the LCM matches b exactly—implying a divides b . In that case, b would invalidly represent the full number itself rather than a proper divisor. To resolve this cleanly, I scale b by the internal ratio b / a , generating the minimal possible valid number that conforms to the target constraint. This yields a highly efficient time complexity of $O(\log(\min(a, b)))$ because it relies fundamentally on standard GCD and LCM calculations."

Strategic Takeaway: Process over Product

Never initiate your solution description by reciting a raw conditional formula like `if (lcm == b) return b * (b / a);`. Starting with a hard-coded formula signals rote memorization rather than raw engineering synthesis.

Instead, guide the interviewer through the logical cascade: **Observation** (→ both numbers divide x) • **Deduction** (→ must be a common multiple) • **Optimization** (→ check the LCM first) • **Edge Case Handling** (→ identifying the identity collision) • **Resolution** (→ proportionate scaling). This framework proves you can derive solutions dynamically under pressure.